

Vinicius Arruda



personal information

26 years old, brazilian

contact

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Itapuã
Vila Velha - ES, Brazil

website

viniciusarruda.github.io

idioms

Portuguese,
English,
German

programming languages

C, Python, MATLAB

skills

Artificial Neural Networks,
Data Analysis, GitHub, $\text{\LaTeX}$,
Linux, Machine Learning,
Metaheuristics, Scientific
Research, TensorFlow, Time
Series Prediction

summary

Undergraduate student of Computer Science at Federal University of Espírito Santo, with experience in artificial neural networks and scientific research. Has as main interest artificial intelligence and neuroscience.

education

2014 - 2018	Computer Science GPA: 9.01/10.0*	Federal University of Espírito Santo, UFES *according to the brazilian criteria
2012 - 2014	Technical Course in Informatics Final Project: A sniffer using the libpcap library and C programming language	Federal Institute of Espírito Santo, IFES
2010 - LOA	Civil Engineering	University of Vila Velha, UVV

experience

3/2018 - 5/2018	Projeta Sistemas <i>Internship</i>	Vitória, Brazil
Implementation and testing of image processing algorithms and convolutional neural networks applied to stereo images of cattle, in order to measure the cattle weight without the use of weight scale. During the internship it was used the programming languages Python and C++ with the libraries OpenCV and TensorFlow.		
5/2015 - 11/2016	Federal University of Espírito Santo <i>Undergraduate Researcher</i>	Vitória, Brazil
Research in artificial neural networks and metaheuristics, applied to time series prediction with focus on the algorithms Extreme Learning Machine and Particle Swarm Optimization. This research resulted in a full paper publication at the XLVIII Brazilian Symposium of Operational Research and was funded by the FAPES research agency. The programming languages used during the research were MATLAB, C, C++, Python and MQL5.		

publications

30/9/2016	Particle Swarm Optimization for training artificial neural networks of type ELM: A case study for time series prediction
XLVIII Brazilian Symposium of Operational Research	
Abstract: The use of Artificial Neural Networks (ANN) for the time series prediction problem has shown promising results. However, standard training methods of ANNs based on gradient as the backpropagation algorithm may get trap into local minima and the learning of the network stagnates. Moreover, the training of ANN with backpropagation may be very time consuming. In order to tackle these issues, the Extreme Learning Machine (ELM) algorithm was developed. Since the ELM algorithm requires the learning of parameters of the network, we present the Particle Swarm Optimization (PSO) algorithm used in conjunction with the ELM algorithm in order to find optimal parameters for the network. In this paper, the hybrid algorithm ELM+PSO is applied to univariate and multivariate and time series prediction problems. The full paper is available at: www.din.uem.br/sbpo/sbpo2016/pdf/155919.pdf	

interests

Artificial General Intelligence, Artificial Intelligence, Artificial Life, Artificial Neural Networks, Bio-inspired Computation, Cognitive Science, Deep Learning, Neuroscience, Research & Development and Visual Cognition.